

Ivonne Berenice Lemus Martínez | Embedded Software Engineer

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Embedded Software Engineer with over 5 years of experience in developing and executing complex embedded software projects. Proficient in **real-time** operating systems (**RTOS**) and **low-level** programming, with a strong knowledge of communication protocols including **CAN**, **I²C**, **SPI**, and **UART**. Demonstrated expertise in utilizing **AUTOSAR** architecture throughout the **automotive** software development life cycle. Experienced in conducting rigorous testing of secure and critical software systems, ensuring reliability and compliance with **automotive** industry standards.

PROFESSIONAL EXPERIENCE

Embedded Software Engineer - BorgWarner. 19-03-2024 – 09-06-2024

- Integrated **pull request** code updates into the project's main branch using **version control tools** such as **Bitbucket**, **Git**, **GitHub**, **Plastic**, and **SmartGit**.
- Collaborated with a **global cross-functional team** to develop customized solutions that resolved customer needs.

Embedded Software Engineer - CIDEA. 16 - 06 2021 - 04-12- 2023

- Designed and implemented a **C-based** software component within the **AUTOSAR** architecture to monitor the ECU circuit board temperature.
- Performed software validation by sending **CAN** messages via the **UDS** (Unified Diagnostic Services protocol) to a test bench and evaluating diagnostic feedback to verify alignment with the detailed software specifications.
- Implemented **software development** in accordance with the V-Model of the **ASPICE (Automotive SPICE)** process to fulfill the **Software Development Life Cycle**.
- Configured 20+ signals parameters according with the standard and **Ethernet (TC/IP)** protocols in the **COM** Module of the Basic Software layer (BSW).
- Improved **ECU** reliability by resolving over 10 code **issues** using analysis and **debugging** tools such as **Ozone** and the **JTAG** interface to identify root causes.
- Provided mentorship to engineers in performing low-level testing, **unit testing**, and static analysis for **safety-critical** systems.
- Developed and implemented a **state machine** to manage the horn feature output in response to CAN signal inputs within the ECU (Electronic Unit Control), as part of an automotive hardware system.
- Performed software **unit tests** with VectorCAST to verify the detailed software design, ensuring high-quality software delivery and full condition coverage of functions.
- Generated code using **MATLAB/Simulink** and integrated the changes into the software layer in accordance with **AUTOSAR** interfaces using Vector DaVinci tools.
- Performed hardware testing to ensure detection of **short-to-battery**, **short-to-ground** conditions in the ECU's power features, while monitoring **UDS** messages and analyzing system responses.
- Identified 10+ code vulnerabilities through static analysis using **LDRA** in Eclipse, reporting MISRA violations, **Secure C** issues, and quality warnings.
- Optimized software development by 80% through the precise specification of software requirements, in accordance with specified guidelines and following the sprint dynamics of the **Agile methodology**.

-Designed software architecture using dynamic and activity **UML** diagrams in Enterprise Architect, increasing coding efficiency by 70%.

Project Engineer - CIDESI. 12-04-2019 -14-06-2021

-Developed software in **Embedded C** language for **ARM microcontrollers** to store data in external NAND flash memory, improving storage speed by 80% in a glider datalogger system.

-Analyzed over 20 analog **schematics** of amplifier circuits to perform reverse engineering using an oscilloscope.

-Implemented code to read data from digital sensors using **Python** and **embedded Linux**, optimizing GPIO (General Purpose Input/Output) usage by 50%.

-Maximized the efficiency of TMS570LS3137 **microcontroller** system up to 30%, by developing **embedded software** implementing **RTOS (Real Time Operating System)**, using **GIT** for program version control.

-Developed **low-level drivers** for **I²C**, **CAN**, **SPI**, & **UART** communication protocols, supporting **debugging** and validation activities, including **troubleshooting** issues.

-Visited customer and internal locations to provide hands-on support to **systems engineers**, identify **hardware** requirements, and contribute to the digitalization of the existing railway driving system in Mexico City.

-Enhanced the Mexico City railway software system by applying **MISRA-C** and **Secure-C** coding standards, resulting in up to a 90% increase in peripheral **communication security** and reliability.

-Performing **software testing** on train tracks and troubleshooting during system start-up to verify correct implementation of the system update.

-Improved the Mexico City railway control system by developing and deploying speed and acceleration control algorithms on **SafeRTOS** and **FreeRTOS** platforms, achieving up to a 90% improvement in system reliability.

Graduate Studies- Instrumented equipment Department - CIDESI. 27-09-2017 – 15-12-2020

-Conditioned signals from 16 Hall sensors for acquisition by an 8-bit resolution ADC by amplifying the original signals up to 10 times.

Internship - Instrumentation Equipment Department - CIDESI. 27- 04-2017 - 15-09-2017

B.S. in Mechatronics Engineering | 09 -02-2012 - 10-02-2017

National Technological Institute of Mexico | San Luis Potosí

-Implemented a virtual instrumentation array applying least squares method to Model del Dynamic control of a DC control using **MATLAB-SIMULINK** and LabVIEW.

EDUCATION

B.S. in Mechatronics Engineering | 09 -02-2012 - 10-02-2017

National Technological Institute of Mexico | San Luis Potosí

-Developed **firmware** for Texas Instruments, Microchip, and ST **microcontrollers** with **Assembly** and **C** languages.

M.Sc. & T. Mechatronics | 07-09-2017 - 09-12-2019

CIDESI | Querétaro

-Created an electronic **architecture** for signal acquisition of 80 Hall sensors implemented in **FPGA**, which allowed to get 400 μ s in acquisition time and reduce by 70% the implementation costs.

SKILLS

Programming Languages: C, C++ & Assembler

DevOps: Git & Plastic

CPU architectures: FPGA (Artix-7), FPGA (Spartan 3E), TMS570LS3137, MSP430F2618, PIC4550, STL microcontroller.

Communication protocols: I2C, SPI, UART & CAN

Operating systems: RTOS, SafeRTOS & FreeRTOS

Languages: English & Spanish

COURSES

C++: UDEMY. 2024-09-10 - 2024-10-10

Vector tools: CIDECE. 2021-06-16 - 2021-09-27